

Experience: Large Scale Indoor Location-based Service in Libraries

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ABSTRACT

This paper presents the experience in the implementation and application of a large-scale Indoor Location-Based Service (LBS) in an academic library. We deployed an indoor positioning system within the school library that leverages 550 Bluetooth beacons, covering an area of approximately 12,000 square meters. This system allows users to engage with location-aware book navigation services via a web interface on their mobile devices. Upon locating a book of interest, the system enters navigation mode, updating and guiding users based on their current location within the library. Utilizing the users' positional data and borrowing history, the system is capable of recommending other potentially interesting books, exhibits, and library events. The system went live in February 2023, with a recorded usage sessions of 21,540 instances which includes 17,271 uses of the retrieval services and 2,065 indoor navigation services, as of May 2023. This paper outlines the system architecture of this large-scale indoor LBS, shares user engagement data, and after anonymization, makes this data publicly available for academic analysis and research. We are hoping that our experience can shed light on the understanding and future development of large-scale indoor LBS technologies.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in ubiquitous and mobile computing**.

KEYWORDS

location-based service, library

ACM Reference Format:

Ling Ma, Runting Zhang, and Xiaohua Shi. 2023. Experience: Large Scale Indoor Location-based Service in Libraries. In *Adjunct Proceedings of the 2023 ACM International Joint Conference on Pervasive and Ubiquitous Computing & the 2023 ACM International Symposium on Wearable Computing (UbiComp/ISWC '23 Adjunct)*, October 08–12, 2023, Cancun, Quintana Roo, Mexico. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3594739.3610725>

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<https://doi.org/10.1145/3594739.3610725>

1 INTRODUCTION

Motivation: Indoor Location-Based Services (LBS) [2] offer a multitude of opportunities for enriching user experiences in various settings. These services include wayfinding in large indoor spaces such as malls or airports, targeted advertising, inventory management, emergency services, and resource tracking, among others. The relevance of studying and implementing Indoor LBS is three-fold. First, as urban spaces grow more complex, aiding users in navigating these indoor environments becomes increasingly essential. Second, precise indoor localization can provide context-aware services that enhance user interaction and engagement within a specific setting, improving overall user experience. Finally, the extensive data generated through indoor LBS can support decision-making processes, enabling optimization of resources and space management.

As a critical foundation of Indoor Location-Based Services, indoor localization technology has undergone extensive research and development over the years. Various methodologies have emerged, each boasting its unique strengths and wrestling with particular limitations. These include: i) **Infrared-based systems** [1] relying on line-of-sight communication between emitters and receivers offer high accuracy but limited by physical obstructions. ii) **Ultrasonic-based systems** [17] Offer decent precision through the use of diverse signals and information types, such as pulse, FMCW, and phase, these systems are, however, susceptible to environmental noise and constrained by limited bandwidth. iii) **Wireless-based systems** [13] are broadly studied in recent years, technologies such as RFID, WiFi, Bluetooth, UWB, mmWave, etc., display different advantages under various scenarios. Among these, WiFi and Bluetooth-based localization technologies have garnered significant attention due to their widespread availability in many commercial off-the-shelf (COTS) mobile devices. iv) **Visible Light Communication (VLC)** [19] offers high positional accuracy but require line-of-sight. The choice of technology largely depends on the specific requirements of the application, including considerations like system cost, setup complexity, accuracy needs, infrastructure constraints, and user privacy [7, 14, 20, 23].

While indoor localization technologies have been researched extensively for many years, and a variety of techniques have been developed, examples of large-scale deployments and practical applications are still relatively rare [6]. Several reasons account for this. Firstly, the cost and complexity of setting up comprehensive

indoor localization systems can be prohibitive, especially when they require significant modifications to the existing infrastructure. Secondly, achieving high positioning accuracy in real-world scenarios is still a challenging task due to environmental factors such as signal interference, multipath propagation, and dynamic changes in the indoor environment. Thirdly, there is often a trade-off between user privacy and the level of service that indoor localization can provide, which needs to be carefully managed.

Furthermore, there is a lack of available open datasets for academic study in this field, which hinders further technological advancements. These datasets, which provide real-world usage statistics and behavior patterns, are invaluable for researchers aiming to refine and optimize indoor localization systems. By sharing our experiences and anonymized data collected from our large-scale indoor LBS system in an academic library, we hope to shed light on future research and contribute to the advancement of indoor LBS technologies.

Our Application: University libraries face the challenge of transforming their services towards intelligent services [18], a transformation underpinned by five key elements: resources, technology, services, librarians, and users [3]. The necessity of this transformation is rooted in the quest to build a perceptive, personalized, and on-demand service system capable of adapting to the continuously evolving and increasing user demands. By harnessing the power of indoor LBS, we have ventured into the development of a smart library at our institution [15].

The application of our indoor LBS in the academic library [21] stands as a response to the need for modernization and improvement of library services. With the capability to provide real-time location-based information to users, it significantly enhances the library's operational efficiency and improves the book searching experience for readers. Through this real-world implementation, we offer a practical glimpse into the potential of indoor LBS, thus setting a precedent for future large-scale indoor LBS applications.

The primary challenge in constructing a smart library [16] lies in fully leveraging new technologies to empower spatial data and to enable intelligent applications. The intricacies of integrating diverse technologies into an existing system and the need to ensure optimal user experience present significant hurdles. To enhance library operational efficiency and the efficiency of book searches for readers, we proposed and implemented an indoor positioning and navigation system. This system capitalizes on the high-speed transmission capability of 5G technology [9], the user-friendly nature of microservices architecture [25], and the precision of modern positioning technology, thus providing robust technical support for library services.

Despite these advancements, the journey to build a smart library is not without its difficulties. Balancing between cost and efficiency, ensuring privacy while providing personalized services, and maintaining system robustness amidst changes in the indoor environment pose ongoing challenges. However, our system, through its scalable architecture and thoughtful design, demonstrates a promising way forward.

Through rigorous study, analysis, and optimization, we've successfully developed an intelligent library application based on our indoor positioning and navigation system. The system integrates

data from the inventory service platform [10] to provide real-time shelf information and inventory services, thus enhancing the efficiency and convenience of library management. It provides users with a seamless library experience by enabling them to quickly locate needed books and offers accurate personal positioning, spatial positioning, and book information services.

Our system, in essence, has set the stage for the evolution of modern library services by demonstrating the power of large-scale indoor LBS in a practical setting. It also provides fresh insights and methods for the future of intelligent library services [22], opening up new avenues for researchers and practitioners in this field. Our work is not just a technical accomplishment but a testament to the potential of Indoor Location-Based Services in transforming the landscape of public services.

The rest of the paper is organized as follows: Section 2 dives into the detailed system design, explaining how our indoor positioning and navigation system has been implemented to serve the library. Section 3 introduces the data collection process and offers an analysis of user interaction with our system, providing valuable insights into its performance and usage patterns. Finally, Section 4 concludes the paper and discusses potential future directions for the development and application of large-scale indoor LBS.

2 SYSTEM DESIGN

In designing the intelligent positioning and navigation system, meeting the requirements demanded the construction of a system on the 5G converged campus network [5] positioning environment to ensure data security through 5G converged campus network data transmission. The system leverages Bluetooth environment positioning to locate readers and the inventory service [11] platform to confirm the position of books and bookshelves. The library management system provides support to reliably locate the books and obtain real-time location and bookshelf information. The system directly accesses book and reader data from the library management system and inventory service platform shelf data. The construction of the system also included the development of a positioning solution platform and map management platform to obtain location data and map data. The system provides three essential features: mobile search, virtual library, and book finding, satisfying readers' experience needs while exhibiting scalability and self-adaptability.

2.1 System Architecture

We first outline the architecture design of the intelligent positioning and navigation system, which is characterized by flexibility, real-time capability, precision, and controllability. The architecture diagram is shown in Fig. 1. The architecture is designed to flow from the basic data layer, through various interface systems, and then obtain related data from the associated information layer. The technical layer introduces various technologies and provides function virtualization for diversified mobile applications at the application layer. That is, each layer is designed to meet specific goals and requirements:

- **Fundational Data Layer:** To provide a foundation for the intelligent positioning and navigation system, the basic data layer integrates 5G network data, positioning terminal data, and reader data.

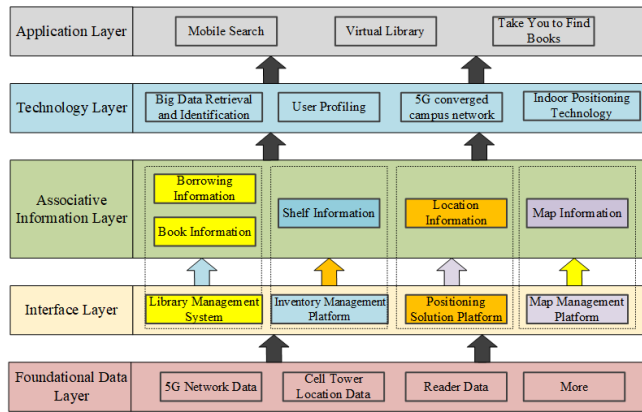


Figure 1: The architecture diagram of the intelligent positioning and navigation system for libraries.

- **Interface Layer:** To cater to users' needs and provide necessary services for the associated information layer, the interface layer includes the library management system, inventory service platform, and map management platform.
- **Associated Information Layer:** This layer plays an essential role in the intelligent positioning and navigation system by collecting and processing important information from the interface layer, including borrowing information, book information, shelf information, and map information. These data types are crucial for the intelligent positioning and navigation system's functioning.
- **Technology Layer:** At the technical layer, we have employed several innovative technologies such as 5G converged campus network [4], big data retrieval and recognition, user profiling [8], and 5G indoor positioning technology to provide robust technical support for the implementation of the application layer.
- **Application Layer:** We integrated functions such as search, borrowing, positioning, and navigation into mobile applications. Multiple functional modules such as virtual library, mobile search, and book finding have been realized to provide users with integrated book search and positioning services.

2.2 Usage Flow Design

This article introduces a service process based on the H5 application format that is embedded into our university's official platform [26] in a modular fashion. The module not only demonstrates but also provides the functionality of the intelligent positioning and navigation system through the following four processes: "Mobile Search", "Virtual Library", "Book Finding", and "My Information". In *Mobile Search*, users can enter keywords to search for books and find the specific location of the book from the results. In *Virtual Library*, users can browse various areas of the library, check bookshelf and book information, and use the positioning and navigation system to find the location of the book. In *Book Finding*, users can enter the name of the target book, and the system will display the location of the book and provide navigation services. In *My Information*, users can view personal borrowing information, search history, and navigation records, etc.

These design processes simplify the task of finding books using the intelligent positioning and navigation system with the help of



Figure 2: The user interface for "Take You to Find Books".

intuitive and clear operation steps, while also conveniently managing personal borrowing information.

2.3 Functions Design

We developed three core modules for our application service: "Mobile Search", "Virtual Library", and the key feature "Take You to Find Books". During the design process, we have emphasized the retrieval module as the core of our main interface, which comprises three modules and a set of four fields dedicated to "My Information". By using our university's official management system, we can authenticate users and integrate their identity information with the library management system to retrieve users' circulation information. Concurrently, our system connects with additional platforms to acquire real-time bookshelf data, maps, and user location data to support the overall user experience.

Mobile Search: The mobile search module allows users to conduct single and advanced searches by entering keywords or search terms. We have developed a classification function that sorts on-shelf books into the 22 categories provided by the Chinese Library Classification, ensuring that user search results are more accurate. Additionally, book recommendation features such as borrowing rankings, search rankings, and new book recommendations are available, providing users with better access to information about the library's resources and latest books on the shelves.

Virtual Library: The virtual library is a digital platform that stores information resources on network carriers. This enables users to search and navigate the library at any time and from anywhere, making it an essential component for developing a digital twin library. The virtual library functions as a guide for physical books and devices, and provides users with access to information about study rooms and equipment. Additionally, it offers navigational support for books, equipment, and study rooms within the library, further enhancing the user experience.

Take You to Find Books: The "Take You to Find Books" module is the primary function of the smart positioning navigation system. The interface is shown in Fig. 2. It seamlessly integrates various platforms such as the library management system, inventory service platform, positioning settlement system and map management

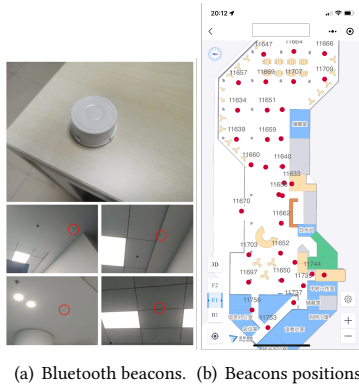


Figure 3: (a) Battery-powered Bluetooth beacons are installed in the ceiling, periodically broadcasting beacon messages. (b) The interface depicts the distribution and density of the Bluetooth beacons in the facility.

platform, to enhance the overall user experience. This module combines book retrieval with user positioning, marking a new milestone for the library in the metaverse [12].

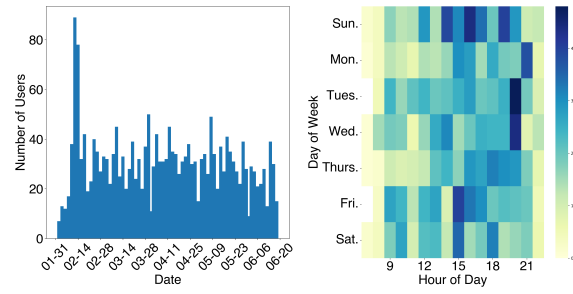
In this module, we leverage the benefits of Bluetooth iBeacon [24] positioning technology and 5G data transmission capabilities. The accurate user positioning is achieved with iBeacon technology, while the fast and stable data transmission is ensured by 5G. This allows us to offer comprehensive, real-time services that combine search, map navigation, and positioning.

We have incorporated Bluetooth iBeacon positioning technology to accurately determine the user's position and offer real-time shelf position information for library books. This enables precise navigation services. Additionally, we have developed a positioning algorithm that utilizes Bluetooth iBeacon positioning technology to optimize the route planning algorithm. This reduces the deviation of the positioning coordinates, ensuring smoother navigation.

Furthermore, the high-speed data transmission capability of 5G enables us to offer high-precision and high-speed book search and positioning navigation functionality. This module not only incorporates mobile search and virtual library functions, but more importantly, it enables users to search and navigate the library using their mobile devices from anywhere at any time, thereby achieving the library's digital twin. This design reflects a seamless integration of Bluetooth iBeacon positioning technology with 5G data transmission capabilities, significantly enhancing the overall user experience and library service efficiency.

3 DATA COLLECTION AND ANALYSIS

Based on our system architecture design and application process introduced in Sec. 2, we have developed a front-end application and a back-end management platform. The platform is connected to positioning calculation servers and map management servers, operating within the 5G network's coverage. The system relies on Bluetooth iBeacon positioning technology which offers highly accurate and reliable positioning data, allowing for precise indoor positioning services. Currently, the library has completed the first phase of the positioning base station and testing environment deployment, and is moving onto the second phase of the implementation and application development processes as per the smart positioning



(a) Number of navigation usage. (b) Usage pattern across days and weeks.

Figure 4: Our smart library system have been recording all of the usage data. The data will be available as soon as the manuscript is published.

navigation plan. Our ultimate objective is to fully-utilize the smart positioning navigation system in the library.

The implementation process involved the employment of 5G transmission and Bluetooth iBeacon technology to enable real-time positioning and to provide a solid framework that can seamlessly support a future transition to pure 5G positioning. As shown in Fig. 3, we deployed an indoor positioning system within the school library that leverages 550 Bluetooth beacons, covering an area of approximately 12,000 square meters.

As of May 2023, our smart library system has recorded 21,540 visitor sessions, including a cumulative total of 17,271 uses of the retrieval services and 2,065 uses of the positioning navigation services. Fig. 4 shows the number of navigations services over time and the usage pattern across days and weeks. These statistics provide strong evidence of the positive impact of our smart positioning navigation system on the quality and efficiency of library services. The adoption of Bluetooth iBeacon positioning technology has significantly enhanced the accuracy of the positioning feature, contributing to the increased effectiveness of the system.

4 CONCLUSION

After extensive technical research and detailed analysis, this study has successfully developed and implemented a distinctive intelligent positioning and navigation system for libraries. The system integrates the positioning and navigation requirements, Bluetooth iBeacon positioning technology, and advanced 5G data transmission technology. We have outlined the system's overall architecture, precise positioning scheme, and diversified application design in great detail. Additionally, we have evaluated the system's effectiveness and the value it contributes to library services.

In conclusion, the intelligent positioning and navigation system described in this paper offers a new library service model that incorporates Bluetooth iBeacon positioning technology and 5G data transmission technology. The system brings with it significant practical value and theoretical implications. Our aim in future research and practice is to further optimize and enhance this model in order to better serve libraries and their readers.

ACKNOWLEDGMENTS

This work was supported by the National Social Science Foundation President Project (Grant No. 20FTQB012).

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